

Module Description

MW2383: Design and Analysis of Digital Control Systems

Chair of Cyber-Physical Systems in Production Engineering
(Prof. Caccamo)

Module level: Bachelor	Language: English	Module duration: one semester	Occurrence: winter semester
Credits*: 5	Total number of hours: 150	Self-study hours: 90	Contact hours: 60

* The number of credits can vary depending on the corresponding SPO version. The valid number is always indicated on the Transcript of Records or the Performance Record.

Description of achievement and assessment methods:

The module will be evaluated via an exercise of 4-6 assignments during the semester. The exact number of assignments will be announced at the start of the lecture. The assignments cover topics for the programming of digital control systems. The final grade will be calculated based on the sum of points collected in these assignments. The students' assignment solutions are to be presented in the tutorial classes.

During the semester, students work on programming tasks that demonstrate that they can design and analyze time-sensitive software for cyber-physical applications and use low-level hardware features such as hardware timers, interrupts, pulse wide modulation (PWM), analog-to-digital, and digital-to-analog converters for microprocessor programming. This type of verification is necessary because the desired learning result can only be achieved through this practical application of methods and basic concepts (programming).

Possibility of re-taking:

In the next semester: Yes
At the end of the semester: No

(Recommended) requirements:

The basic mathematics module of the first year for TUM-BWL students is a prerequisite. It is helpful to attend IN8005 Introduction into Computer Science (for non-Informatics studies, TUM BWL).

Contents:

This course provides an introduction to C programming, embedded systems, and control engineering.

The course is structured along these three main topics:

1. C Programming and coding structure
 - a. Programming languages, compilation and compilers
 - b. Errors and debugging
 - c. C syntax, variables, data types and operators

- d. Flow control statements
- e. Pointers and arrays
- f. Type conversion and casting
- g. Lifetime and scope of variables
- h. Input and output libraries, file handling
- 2. Embedded systems for digital control systems: Arduino board
 - a. Embedded systems introduction
 - b. Designing programs for the Arduino platform
 - c. Digital and analog inputs and outputs
 - d. Reading sensors and controlling actuators
 - e. Time- and event-controlled systems (interrupts and polling)
- 3. Introduction to control engineering
 - a. System description and modeling
 - b. Mathematical background: differential equations, Laplace transform
 - c. Modeling and simulation with MATLAB/Simulink
 - d. Proportional, integral and derivative control
 - e. System stability and controller performance

Study goals:

Upon successful completion of the module, the participants understand important concepts and methods required to model and analyze modern digital control systems. The participants are able to write simple programs in C independently and have an understanding of data types and program structure. They understand the importance of embedded systems for digital control systems and can develop software interacting with the physical environment. The students can apply the basic control engineering methods and simulate and analyze simple systems in MATLAB/Simulink.

Teaching and learning methods:

The module consists of a lecture and tutorial classes. The lecture will introduce new concepts, illustrated and supported with slides and live programming examples. This approach helps to impart basic knowledge and allows students to understand different methods to model and analyze modern digital control systems. Students will apply and deepen their newly acquired knowledge through 4-6 project-oriented assignments in groups to demonstrate the practical relevance of the concepts introduced and explained during the lecture. Solving the assignments is supported by the tutorial classes, allowing students to interact with experienced teaching staff directly.

Media formats:

Presentation slides, annotations made during the lecture, examples of solutions, and exercises templates.

Literature:

- Kernighan, Brian W., and Dennis M. Ritchie. "The C programming language." (2002).
- Kurniawan, Agus. "Getting Started with Matlab Simulink and Arduino." (2013).

Responsible for the module:

Caccamo, Marco; Prof. Dr.: mcaccamo@tum.de

Courses (Type, SH) Lecturer:

0000002158 Design and Analysis of Digital Control Systems (2SWS VO, WS 2024/25)
Cao H, Trumpp R, Caccamo M

0000003336 Tutorials on Design and Analysis of Digital Control Systems (2SWS UE, WS 2024/25)
Cao H, Trumpp R, Dirnberger L, Sun B

For further information about this module and its allocation to the curriculum see:
<https://campus.tum.de/tumonline/wbModHb.wbShowMHBReadOnly?pKnotenNr=1495705>

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